=== UNIT 2: LASER PHYSICS, CAVITY OPTICS & APPLICATIONS ===

>> 2.1 Interaction of Radiation with Matter

LASER is an acronym for Light Amplification by Stimulated Emission of Radiation. Lasers emit coherent, monochromatic, highly directional, and intense optical radiation.
* Three Fundamental Quantum Transitions: Consider two quantized atomic energy levels E1 (ground state) and E2 (excited state), separated by energy Delta E = E2 - E1 = h*nu:
- 1. Induced / Stimulated Absorption: An atom in ground state E1 absorbs an incident photon of energy h*nu and transitions to excited state E2. Transition rate: R_abs = B12 * N1 * u(nu), where u(nu) is radiation energy density.
- 2. Spontaneous Emission: An atom in excited state E2 transitions spontaneously to E1 without external triggering, releasing a photon of energy h*nu in a random direction and random phase. Transition rate: R_spont = A21 * N2.
- 3. Stimulated Emission: An incident trigger photon of energy h*nu interacts with an excited atom at E2, inducing it to drop to E1 and emit a SECOND photon that is identical in frequency, phase, polarization, and direction of propagation. Transition rate: R_stim = B21 * N2 * u(nu).

>> 2.2 Einstein’s A and B Coefficients & Thermodynamic Equilibrium

Albert Einstein (1917) derived the fundamental mathematical relations governing laser action by considering matter and radiation in thermal equilibrium inside an isothermal blackbody enclosure at temperature T:
* Boltzmann Population Ratio: N2 / N1 = e^(- (E2 - E1) / (k_B * T)) = e^(- h*nu / (k_B * T)).
* Balance of Absorption and Total Emission: B12 * N1 * u(nu) = A21 * N2 + B21 * N2 * u(nu).
* Solving for Energy Density u(nu): u(nu) = A21 / [(B12 * N1 / N2) - B21] = (A21 / B21) / [(B12 / B21) * e^(h*nu / k_B*T) - 1].
* Comparison with Planck’s Blackbody Radiation Law: u(nu) = (8 * pi * h * nu^3 / c^3) / [e^(h*nu / k_B*T) - 1].
* Einstein Relations: (1) B12 = B21 (Probability of stimulated absorption equals probability of stimulated emission).
(2) A21 / B21 = (8 * pi * h * nu^3) / c^3. The ratio of spontaneous to stimulated emission is proportional to the cube of optical frequency (nu^3).

>> 2.3 Population Inversion and Pumping Mechanisms

In thermal equilibrium, Boltzmann statistics dictates N2 << N1 (absorption dominates, light is attenuated).

* Population Inversion: A non-equilibrium state where the population of the upper excited state exceeds the ground state: N2 > N1. Light passing through the medium is amplified exponentially: I(x) = I_0 * e^(gamma * x).
* Metastable State: An atomic excited state having an exceptionally long radiative lifetime (tau ~ 10^-3 to 10^-2 s, compared to ordinary states with tau ~ 10^-8 s). Acts as a population storage reservoir.
* Pumping Methods: External energy delivery to achieve population inversion: (1) Optical Pumping (flashlamps for Ruby and Nd:YAG), (2) Electrical Discharge / Electron Collision (He-Ne and CO2 lasers), (3) Direct Carrier Injection (Semiconductor diode lasers).

>> 2.4 Optical Resonant Cavities and Threshold Condition

An optical resonator consists of a pair of aligned mirrors (one 100% reflecting, one partially transmitting ~95-99% reflecting) enclosing the active gain medium:
* Threshold Gain Coefficient (g_th): Amplification per round-trip must balance internal absorption losses (alpha_s) and output mirror transmission losses:
Round-trip gain condition: R1 * R2 * e^((2 * g_th - 2 * alpha_s) * L) = 1.
Formula: g_th = alpha_s + (1 / (2 * L)) * ln(1 / (R1 * R2)).

>> 2.5 Engineering Construction of Important Laser Systems

* 1. Ruby Laser: Three-level solid-state laser. Active medium is a synthetic corundum crystal rod (Al2O3 doped with 0.05% Cr3+ ions). Optical pumping via helical xenon flashlamp (green 550 nm and violet 410 nm bands). Transitions from 2E metastable level to ground state produce pulsed deep red laser output at lambda = 694.3 nm.
* 2. Helium-Neon (He-Ne) Laser: Four-level continuous-wave gas laser. Gas mixture 10:1 ratio of He to Ne at ~1 Torr inside quartz tube. Electrical discharge excites He atoms to 2^1S and 2^3S metastable levels via electron impact. Resonant resonant collision transfers energy to identical energy levels of Ne (3s and 2s). Radiative cascade from 3s2 to 2p4 yields stable continuous red laser light at lambda = 632.8 nm.
* 3. Semiconductor Diode Laser: Heavily doped p-n junction (e.g. GaAs / AlGaAs). Forward biasing injects high concentrations of electrons and holes into the active depletion layer, where direct band-to-band recombination emits stimulated coherent infrared/optical radiation.

>> 2.6 Civil and Surveying Engineering Applications of Lasers

* Laser Distance Measurement (EDM & Total Station): Measures time-of-flight or phase shift of modulated laser beams for millimeter-accuracy geodetic surveying.
* LIDAR (Light Detection and Ranging): Airborne and terrestrial 3D laser scanning generates high-density digital elevation models (DEM) for highway corridors,

landslides, and flood modeling.
* Tunneling and High-Rise Construction Alignment: Narrow laser beams guide tunnel boring machines (TBM) over multi-kilometer underground drives with sub-centimeter alignment accuracy.
* Structural Health Monitoring: Laser Doppler vibrometry non-contact measurement of dynamic vibrations and modal frequencies of long-span bridges.

[EXAMPLE 1: Ratio of Spontaneous to Stimulated Emission at Room Temperature]

Problem Context: An optical transition produces visible red light at wavelength lambda = 632.8 nm (He-Ne laser). Calculate the ratio of spontaneous emission rate to stimulated emission rate at room temperature (T = 300 K). State why population inversion cannot occur in thermal equilibrium.
Solution Step 1: Compute Transition Frequency nu and Photon Energy h*nu:
nu = c / lambda = (3.0 * 10^8 m/s) / (632.8 * 10^-9 m) = 4.741 * 10^14 Hz.
Photon energy Delta E = h * nu = (6.626 * 10^-34 J-s) * (4.741 * 10^14 s^-1) = 3.141 * 10^-19 J.
In electron-volts: Delta E = 3.141 * 10^-19 / (1.602 * 10^-19) = 1.961 eV.
Solution Step 2: Compute Thermal Energy k_B * T at 300 K:
k_B * T = (1.381 * 10^-23 J/K) * 300 K = 4.143 * 10^-21 J = 0.02586 eV (~ 26 meV).
Solution Step 3: Compute the Ratio (R_spont / R_stim):
R_spont / R_stim = A21 / (B21 * u(nu)) = e^(h*nu / k_B*T) - 1.
Exponent: h*nu / (k_B*T) = 1.961 eV / 0.02586 eV = 75.83.
R_spont / R_stim = e^(75.83) - 1 ~= 8.56 * 10^32.
Conclusion: Spontaneous emission exceeds stimulated emission by 10^32 times at thermal equilibrium! Stimulated emission is completely negligible under normal thermal conditions, proving that lasers require active non-equilibrium pumping.

[EXAMPLE 2: Threshold Gain Coefficient for a Laser Cavity]

Problem Context: A semiconductor laser has an active cavity length L = 500 micrometers (0.50 mm). The internal scattering loss coefficient is alpha_s = 10 cm^-1 (1000 m^-1). The cleaved end facets act as mirrors with reflection coefficients R1 = R2 = 0.32. Calculate: (1) Threshold gain coefficient g_th, (2) If length is doubled to L = 1000 micrometers, calculate the new threshold gain.
Solution Step 1: Formula for Threshold Gain:
g_th = alpha_s + (1 / (2 * L)) * ln(1 / (R1 * R2)).
Mirror product R1 * R2 = 0.32 * 0.32 = 0.1024.
ln(1 / 0.1024) = ln(9.7656) = 2.2789.
Solution Step 2: Calculate g_th for L = 500 micrometers (0.050 cm):
Mirror loss term = (1 / (2 * 0.050 cm)) * 2.2789 = 10.0 * 2.2789 = 22.79 cm^-1.
g_th = 10.0 + 22.79 = 32.79 cm^-1 (3279 m^-1).

Solution Step 3: Calculate g_th for L = 1000 micrometers (0.10 cm):
Mirror loss term = (1 / (2 * 0.10 cm)) * 2.2789 = 5.0 * 2.2789 = 11.39 cm^-1.
g_th_new = 10.0 + 11.39 = 21.39 cm^-1 (2139 m^-1).
Conclusion: Doubling cavity length reduces mirror transmission losses per unit length, lowering threshold gain requirement from 32.79 cm^-1 to 21.39 cm^-1.

[EXAMPLE 3: He-Ne Laser Beam Divergence and Spot Size on Target]

Problem Context: A civil engineering tunnel alignment He-Ne laser (lambda = 632.8 nm) emits a Gaussian laser beam with an initial beam waist diameter 2*w_0 = 1.2 mm and power P = 5.0 mW. Calculate: (1) Beam divergence half-angle theta_div, (2) Beam spot diameter at a distance D = 2.5 km down a tunnel, (3) Optical power density (intensity) at the target.
Solution Step 1: Beam Waist Radius and Divergence Angle:
Waist radius w_0 = 1.2 mm / 2 = 0.60 mm = 6.0 * 10^-4 m.
Divergence half-angle (diffraction limit): theta_div = lambda / (pi * w_0) = (632.8 * 10^-9 m) / (pi * 6.0 * 10^-4 m) = 3.357 * 10^-4 radians (0.336 mrad).
Solution Step 2: Spot Diameter at Distance D = 2500 m:
Beam spot radius at distance D: w(D) ~= theta_div * D = (3.357 * 10^-4 rad) * 2500 m = 0.8393 m.
Spot diameter 2*w(D) = 2 * 0.8393 = 1.679 m (~ 1.68 m).
Solution Step 3: Optical Power Density at Target:
Cross-sectional spot area A = pi * (w(D))^2 = pi * (0.8393)^2 = 2.213 m^2.
Average Intensity I = Power / Area = 5.0 * 10^-3 W / 2.213 m^2 = 2.26 * 10^-3 W/m^2 = 2.26 mW/m^2.
Conclusion: Divergence angle = 0.336 mrad; Beam spot diameter at 2.5 km = 1.68 m; Target intensity = 2.26 mW/m^2.

[EXAMPLE 4: Temporal Coherence and Coherence Length of Laser Diode]

Problem Context: A distributed feedback (DFB) laser diode used in fiber optic structural strain sensing has a center emission wavelength lambda_0 = 1550 nm and a narrow spectral line width Delta lambda = 0.02 nm. Calculate: (1) Spectral frequency bandwidth Delta nu, (2) Coherence time tau_c, (3) Coherence length L_c in free space.
Solution Step 1: Frequency Bandwidth Delta nu:
Relation between wavelength spread and frequency spread: Delta nu = (c / lambda_0^2) * Delta lambda.
Delta nu = [(3.0 * 10^8 m/s) / (1550 * 10^-9 m)^2] * (0.02 * 10^-9 m).
Delta nu = (3.0 * 10^8 / 2.4025 * 10^-12) * 2.0 * 10^-11 = 1.2487 * 10^20 * 2.0 * 10^-11 = 2.497 * 10^9 Hz = 2.50 GHz.
Solution Step 2: Coherence Time tau_c:
tau_c = 1 / Delta nu = 1 / (2.497 * 10^9 s^-1) = 4.004 * 10^-10 s = 0.40 nanoseconds.

Solution Step 3: Coherence Length L_c:
L_c = c * tau_c = c / Delta nu = lambda_0^2 / Delta lambda.
L_c = (1550 * 10^-9)^2 / (0.02 * 10^-9) = 2.4025 * 10^-12 / 2.0 * 10^-11 = 0.1201 m = 12.01 cm.
Conclusion: Frequency bandwidth = 2.50 GHz; Coherence time = 0.40 ns; Coherence length L_c = 12.01 cm.

>> 2.7 University Review & Examination Questions

* Q1: Derive Einstein's relations for radiation-matter interaction and explain why stimulated emission dominates over spontaneous emission in lasers.
Answer: Under thermal equilibrium at temperature T, the rate of upward transitions (absorption) must equal downward transitions (spontaneous + stimulated emission): B12*N1*u(nu) = A21*N2 + B21*N2*u(nu). Solving for u(nu) and matching with Planck's formula u(nu) = (8*pi*h*nu^3/c^3) / [e^(h*nu/kT) - 1] yields two Einstein relations: (1) B12 = B21 (stimulated absorption and emission coefficients are identical). (2) A21/B21 = 8*pi*h*nu^3/c^3. In thermal equilibrium, A21/(B21*u(nu)) = e^(h*nu/kT) - 1 >> 1 for optical frequencies, meaning spontaneous emission completely drowns stimulated emission. In a laser, external pumping creates population inversion (N2 >> N1) and high resonant cavity photon density u(nu), making stimulated emission overwhelmingly dominant.
* Q2: Explain the construction and four-level working principle of the Helium-Neon (He-Ne) gas laser with an energy level diagram.
Answer: Construction: A He-Ne laser comprises a narrow bore quartz discharge tube filled with helium and neon gas in a 10:1 ratio at low pressure (~1 mm Hg). Electrodes ignite an electrical glow discharge. Two optical cavity mirrors (one 100% reflector, one ~99% output coupler) terminate the tube. Working Principle: Electrons in discharge collide with ground-state Helium atoms, exciting them to long-lived 2^1S and 2^3S metastable levels. Because neon has 3s and 2s levels of nearly identical energy (within 0.05 eV), resonant energy transfer occurs during inelastic atomic collisions: He* + Ne -> He + Ne*. This rapidly pumps neon into 3s states, establishing population inversion relative to the lower 2p states. Stimulated radiative transitions from 3s2 to 2p4 emit the iconic 632.8 nm red laser beam. Neon then drops rapidly to 1s via spontaneous emission and returns to ground state by wall collisions.

>> 2.8 Key Takeaways and Revision Points

* Lasers require three elements: Active gain medium, Pumping source, Optical resonant cavity.
* Population inversion (N2 > N1) requires metastable states and active energy pumping.
* Einstein relation A21/B21 = 8*pi*h*nu^3/c^3 shows spontaneous emission scales with nu^3.
* In civil engineering, lasers enable sub-millimeter precision in LIDAR, Total Stations, and tunneling.